

POKHARA UNIVERSITY

Semester: Fall

Level: Bachelor

Programme: BE Computer

Course: Mobile Apps Development (Elective II)

Year : 2023

Full Marks: 100

Pass Marks : 45

Time : 3hrs

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Define android. Explain different types of mobile operating system in brief. [7]
- b) Differentiate between Native and HTML5 applications with advantages and disadvantages. [8]
2. a) Explain the different data management technique used in android. Explain with code example of any two of them. [8]
- b) Define Sensor. Explain different types of sensors used in mobile application. [7]
3. a) Define Menu. Explain different types of menus used in android apps development. Also give example on any one of them. [7]
- b) Define Activity. Explain the Activity lifecycle in brief. [8]
4. a) Define Intent. How you can switch between multiple Activities explain with code snippet. [7]
- b) Write down the steps to signing and publishing apps to play store. [8]
5. a) Define Layout. Explain different types of layouts used in android apps with example. [8]
- b) What is the importance of Notification in android apps? Illustrate different types of Notification with example. [7]
6. a) Define AsyncTask. When to use it explain with code snippet. [8]
- b) Explain the different types of animation used in android application with example. [7]
7. Write short note (on Any Two) [5x2 = 10]
 - a) Content Provider
 - b) GPS and its uses
 - c) Fragment and Its use

media, message, toast, push, pull
radioes, dialogues, lockscreen

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Image Processing and Pattern Recognitions

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define digital image processing. Explain briefly about Components of an Image Processing System. 8
- b) Apply contrast stretching technique on following 3-bit grey level image of size 4x4. 7

4	1	2	1
7	5	6	5
3	2	2	4
2	2	3	6

2. a) Differentiate between smoothening and sharpening filters. Explain some basic grey level transformations techniques used for image enhancement in spatial domain. 7
- b) Compute the 2-D discrete Walsh-Hadamard transform of the given image block below: 8

5	6	8	10
6	6	6	7
3	5	3	6
7	8	6	7

3. a) Why do we use Haar Transform in Image processing? Explain its algorithm for the order $N=2$. 7
- b) Define salt-and-pepper noise. Which spatial filter is used to reduce salt-and-pepper noise? Why? 8

4. a) What do you mean by lossless image compression? Briefly explain the algorithm for run length coding with suitable example. 8
 b) Find compression ratio and coding efficiency of following 3-bit image after using Huffman Coding technique. 7

Gray level	0	1	2	3	4	5	6	7
No. of Pixel	30	35	38	10	15	10	38	80

5. a) Define the morphological erosion and dilation operators. Which of these can be considered as a noise-reduction technique? Why? 7
 b) Describe how a grayscale image can be segmented using intensity thresholding. You need to explain the cases of single and multiple thresholds. 8
6. a) How can you detect edges with gradient filters? Give different first order derivative based (gradient) filters. 7
 b) For the following image, write down the 8-chain code and find the shape number. I* is the starting point and the rotation is clock-wise direction. 8

		I*	I	I			
	I				I		
	I					I	
	I					I	
		I			I		
			I	I			

7. Write short notes on: (Any two)

2×5

- a) Neural Network
 b) Steps in Pattern Recognition
 c) Image Restoration Vs Image Enhancement

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2023

Programme: BE

Full Marks: 100

Course: Artificial Intelligence

Pass Marks: 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is intelligence? Explain how cognitive science can be used to design a machine which can think like human. 7
 - b) What is a Constraint Satisfaction Problem (CSP)? Solve the cryptarithmic problem: BLACK + GREEN = ORANGE. 8
 2. a) Explain with example how A* search does not stuck in a loop like best-first search and provide a solution if exists. 7
 - b) How does alpha-beta pruning improve the efficiency of minimax search? Explain with example. 8
 3. a) Given the propositional statements p: "It is raining" and q: "The road is wet", write the logical expressions for the following: 8
 - i. If it is not raining, then the road is not wet.
 - ii. If it is raining, then the road is wet, and if the road is wet, then it is raining.
 - iii. It is not the case that both it is raining and the road is wet.
 - iv. If the road is wet, then it is raining, but if it is not raining, then the road is not wet.
 - b) Consider the following facts: 7
 - John is the parent of Bob.
 - Lili is the parent of Bob.
 - John is male.
 - Lili is female.
- Write prolog code to represent these facts and also write rules in prolog to find the father, mother, and children of X. Then make the following query on the knowledge base:
- i. Who is the parent of Bob?
 - ii. Provide a list of all parents and their children.
 - iii. Is John a parent? If so, who are his children?

4. a) How can Explanation-Based learning be used to learn a given object? 8

Explain in detail based on the cup generalization problem.
- b) What is K-means clustering? Cluster the given eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9). 7
5. a) Discuss the role of support vectors? What is the significance of maximizing the margin in SVM? 7
- b) What is a perceptron? Explain how the Adaline network updates its weight during its training. 8
6. a) What is an expert system? How can it help or assist human? Explain with an example. 7
- b) How does ambiguity occur at phonetic, syntactic, semantic and pragmatic levels of natural language processing? Explain with examples. 8
7. Write short notes on: (Any two) 2x5
 - a) Statistical reasoning
 - b) Reinforcement learning
 - c) Hopfield network

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Economics

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss Engineering Economics and describe Principles and applications of Engineering Economics. 7
b) Define manufacturing, non-manufacturing, opportunity and sunk cost with suitable examples. 8
2. a) Mr. Hari Lamsal needs Rs2500000 at the end of 15th year in his bank account. How much money does he need to deposit at the end of every year for 10 years to meet his requirement if his bank provides him 12% interest compounding annually? 8
b) Evaluate the following project under IRR method and also show unrecovered investment balance in tabular and graphical forms. 7
Initial investment: Rs 1500000
Annual revenue: Rs450000
Annual cost: Rs50000
Life: 6 years
Salvage value: 60000
MARR: 13%
3. a) Calculate both types of BC ratios from the following information and also give your decision whether the project should be selected or not. 7
Initial investment: Rs 800000
Annual revenue: Rs 150000
Operating and maintenance cost: Rs25000
Life: 10 years
Salvage value: 20000
MARR: 13%
b) What do you understand by payback period? Calculate simple and discounted payback period from the following information. 8
Initial investment 1000000 MARR 10%

Year	1	2	3	4	5	6
Net Annual Revenue	250000	345000	320000	280000	125000	210000

4. a) Project A and B are mutually exclusive projects. By using repeatability method evaluate the both projects. 8

Items	Project A	Project B
Initial investment	Rs200000	Rs 300000
Annual Revenue	Rs 75000	Rs 85000
Annual Cost	Rs 10000	Rs 12000
Useful life	6 years	8 years
Salvage Value	Rs 50000	Rs 80000

MARR 9% for both projects

- b) Discuss Financial Ratio and explain the followings in detail. 7
 - i. Current Ratio
 - ii. Debt Ratio
 - iii. Total asset Turnover ratio
 - iv. Inventory Turnover ratio
- a) Explain causes of depreciation. A machine costing of Rs. 10,000, estimated Salvage Value of Rs. 500 at the end of 5 year and MARR is 12%. Find depreciation each year by using: 8
 - i. Sinking Fund Factor method.
 - ii. SOYD method.
- b) Pathivara Engineering Works Company has following capital structure. 7

Common Stock Rs 1000000
Bond Rs1500000
Preferred Stock Rs 500000
Total capital Rs 3000000

The company's cost of common stock is 21%
Cost of bond before tax is 18% and tax rate is 40%
Cost of preferred stock is 17%

Calculate weighted average cost of capital (WACC) of the company.
- a) Perform Sensitivity analysis of the following Project over the range of $\pm 30\%$ in i) Initial investment ii) Annual revenue iii) Useful life and iv) Salvage Value: 8

Initial Investment (Rs)	14,500
Annual Revenue (Rs)	4,000
Salvage Value (Rs)	10,000
Useful life (Years)	6
MARR	20%

- b) Prepare Balance sheet of Himalayan Tea Company as on 31st December 2023 from the following transaction. 7

Common Stock	1000000	Land and Building	1362000
Inventories	45000	Cash	75000
Vehicles	25000	Creditors	65000
Bank overdraft	200000	Debtors	120000
Prepaid insurance	25000	Outstanding salary	35000
Marketable securities	28000	Bond and Debenture	350000
Provision for taxation	55000	General Reserves	180000
Furniture	85000	Machinery	120000

7. Write short notes on: (Any two)

2×5

- Nominal and Effective interest rate
- VAT
- Ecological limit and sustainable development

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Network along with its architecture. Explain different kind of services that are benefited by the networking technology in today's world? 4+4
b) Define protocol. Why do we need layered protocol architecture? Discuss each layer of TCP/IP protocol architecture in detail. 7
2. a) Justify the implications of Optical fiber in present day to enable global communications with a suitable example. 8
b) What is Error Detection? Show how CRC technique will help to detect the error with suitable example. 7
3. a) Design an algorithm for CSMA/CD with a suitable example. 8
b) Explain the frame format of IPv4 with a suitable diagram. 7
4. a) Pokhara University is a Member of APNIC. It has acquired an IPv4 pool of (103.32.0.0/22). From the acquired pool, it is required to be subnetted and distributed across Six Schools. Among Six Schools, Two Schools need to be subdivided into two different departments. Design and provide a complete IP Address plan which includes Network Address, Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask for every subnet. 8
b) What is TCP? Explain TCP connection and termination with suitable diagram along with its TCP state while connection and termination. 7
5. a) What is network congestion? Explain how TCP handle congestion. 7
b) Compare Http, Https, document, Html, URL, port in web request. 8

6. a) What do you mean by Asymmetric Cryptography? Write simple out line algorithm in Asymmetric cryptography to generate key, Encryption and Decryption process. 8
b) What is packet Filtering firewall? Show how Differ-Hellman works with help suitable example. 7
7. Write short notes on: (Any two) 2x5
a) Interior and exterior routing
b) Simple Mail Transfer Protocol
c) Framing technique

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: ICT Project Management

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define project. What is ICT Project Management? Which factor is decided the success of project? 8
- b) Why project goals need to be SMART? What kind of environment is considered as favourable for a project? 7
2. a) Describe in detail the Project Integration Management Process. 8
- b) What is WBS? Illustrate a WBS for a project of website design of Pokhara University. 7
3. a) List out few tools used for modeling schedule. Draw Network diagram for the given set of activities, and identify critical path and critical activities. Also calculate the project duration. 8

Activities	Duration	Immediate Successor
P	2	R
Q	3	S
R	4	T, U
S	4	T
T	5	V
U	3	-
V	4	-

- b) What are the processes of communication? Why Communication is important in Project Management? 7
4. a) i. If SPI is 0.95 with earned value being 80. Calculate planned value and also state whether the project is ahead or behind schedule? 8
- ii. A project is scheduled for time of 20 months with an estimated cost of 800000. After 5 months, the evaluation is done and identified that 50% of work is accomplished with cost of 500000. Calculate cost and schedule variance for the project. 7

- b) What are the significance of developing a strong project team? Discuss with examples. 7
5. a) What is risk assessment? Illustrate with an example how risk exposure is calculated in quantitative risk assessment process. 7
- b) What is procurement? What are the objectives of Public Procurement Act in Nepal? 8
6. a) Differentiate Verification and Validation. What are the four processes of project quality management? 7
- b) What is a stakeholder? What kind of stakeholders are there in an ICT project? Explain with examples. 8
7. Write short notes on: (Any two) 2x5
 - a) Organizational Process Assets
 - b) Closing Project
 - c) Forwards and Backwards Pass

